

# Factors associated with oral HPV infection among a large multinational cohort of men

We show that men from the USA, Mexico and Brazil are at risk of acquiring oral human papillomavirus (HPV) throughout their lifetime. Risk factors for new HPV infections are alcohol consumption, having male sexual partners, more lifetime female sexual partners, more oral sex given and higher educational attainment.

## This is a summary of:

Dube Mandishora, R. S. et al. Multinational epidemiological analysis of oral human papillomavirus incidence in 3,137 men. *Nat. Microbiol.* <https://doi.org/10.1038/s41564-024-01824-5> (2024).

## Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Published online: 18 October 2024

## The problem

Human papillomavirus (HPV) is associated with a variety of cancers in men, including oropharyngeal cancer (OPC). The prevalence<sup>1</sup> and incidence of OPC is increasing in the general population worldwide<sup>2</sup>, with the highest incidence occurring in Northern Europe and North America, where new cases occur at a rate of up to 8 per 100,000 men. Despite its high prevalence, however, little is known about the natural history of these infections in OPC. The few studies that have reported on oral HPV acquisition have limited sample size<sup>3</sup>, short follow-up periods<sup>4</sup> or focused on high-risk populations such as men who have sex with men (MSM)<sup>5</sup>. We aimed to estimate the incidence of oral HPV infections and associated risk factors in the HPV In Men (HIM) Study – a large multinational cohort of men from the USA, Mexico and Brazil.

## The solution

To determine the acquisition of oral HPV over time, we carried out a prospective study from November 2007 to September 2016. A total of 3,137 men who provided at least two oral gargle samples, collected every six months, were included in this study, with a median follow-up time of 57 months (range: 6–104 months). HPV genotyping was performed using commercial HPV testing (the SPF<sub>10</sub> PCR-DEIA-LiPA<sub>25</sub> system). Data on sociodemographic characteristics and behavioural history were collected using a computer-assisted self-interview at every visit. An incident oral HPV infection was defined as the first acquired oral HPV genotype that was not present at first visit. Characteristics of the study population were compared across country of residence (USA, Brazil and Mexico) (Fig. 1).

We found that 19.6% of study participants had prevalent oral HPV infection (with any genotype) at baseline. The baseline prevalence of oral oncogenic HPV specifically was 5.9%, with HPV-16 being the most prevalent genotype. For any oral oncogenic HPV, the incidence rate was 2.4 per 1,000 person-months, with higher incidence among men from the USA compared with men from Brazil and Mexico. Risk factors

associated with oral HPV acquisition were alcohol consumption, having male sexual partners, a greater number of lifetime female sexual partners, more oral sex given and higher educational attainment.

## Future directions

Overall, these data suggest that men are at risk of acquiring oral HPV throughout their lifetimes, and identify risk factors associated with oral HPV acquisition. The findings should inform interventions to prevent OPC, such as gender-neutral vaccination and catch-up vaccination of middle-aged men who missed the opportunity to have been vaccinated earlier in life. Awareness of the risks of oral oncogenic HPV acquisition should highlight the association of sexual behaviours and alcohol consumption. The association of longer duration of education with new infections requires further investigation. Given the increasing incidence of HPV-related OPC with increasing age in men, the results of our study highlight the importance of preventing oral HPV, the precursor to HPV-associated OPC.

Several caveats should be considered. First, our research does not allow us to report data after 2016. Second, lifestyle and behavioural variables were self-reported and, as such, reporting bias cannot be ruled out. It should be noted, however, that to minimize this bias the surveys were self-administered via computer in a private room. Finally, the oral rinse and gargle sample collection method does not target the exact cancer site of interest, oropharynx. However, this is the method of choice across multiple studies, as it remains the only way of sampling that area without applying invasive methods. To take this research further, our goal is to investigate the association of longer duration of education with new infections and to guide prevention strategies and awareness campaigns. Furthermore, taking a granular look at the factors that are independently associated with age of acquiring oral HPV is important.

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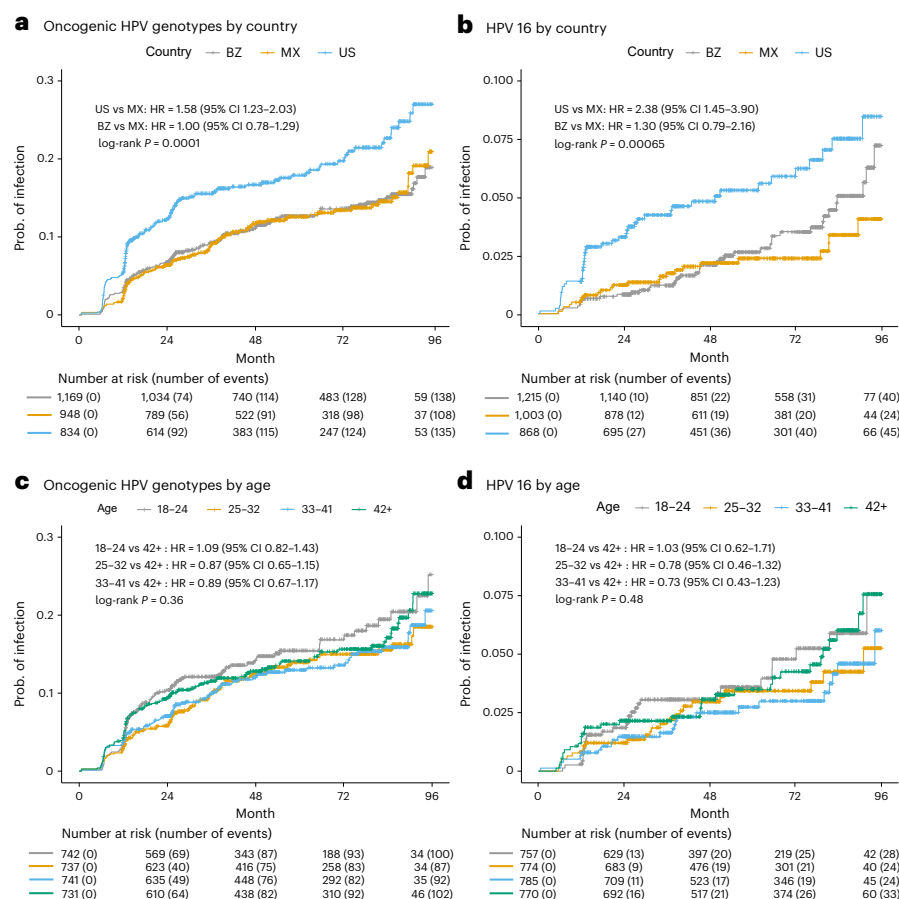
Moffitt Cancer Center, Tampa, FL, USA.

## EXPERT OPINION

"The current study results are of high priority to disseminate, since they will inform modelling studies and public health efforts more thoroughly than the

currently rather patchy data we possess on oral HPV infections over the life span."  
**Karin Sundström, Karolinska Institutet, Huddinge, Sweden.**

## FIGURE



**Fig. 1 | Kaplan–Meier estimates.** a–d, Cumulative incidence of oral oncogenic HPV and HPV-16 infections grouped by country of residence (a,b) and age group (c,d) in a large cohort of men. To test differences among groups, a two-sided  $P < 0.05$  measure of statistical significance log-rank test was conducted, with 95% confidence intervals. Month '0' refers to the first oral gargle sample at baseline. BZ, Brazil; HR, hazards ratio; MX, Mexico; US, USA. © 2024, Dube Mandishora, R. S. et al.

## BEHIND THE PAPER

When we started this project, there were few studies on HPV in men, with all studies at the time focusing on genital HPV. In 2004, we launched the HIM Study, the first prospective study of anogenital HPV among men in the USA, Mexico and Brazil. While the HPV scientific community agreed the study was needed, there was concern men would not participate. In 2005, with sufficient evidence to conclude HPV-16 caused OPC in men, we quickly added

an oral gargle collection to study the natural history of oral HPV. Over the years, we learned the challenges of oral HPV infection studies — namely, that a more sensitive assay would be needed, and that natural history differed by anatomic site and sex. Fast forward 15 years, with almost 3,200 men recruited in the study, we could now initiate the definitive studies of oral HPV natural history. **A.R.G.**

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## FROM THE EDITOR

"This study stood out due to its large cohort, its multi-country aspect and long follow-up period, enabling generalizable conclusions to be drawn on the important health issue of oral oncogenic HPV infection. These conclusions have implications for public health decisions regarding vaccination." **Jess Thompson, Associate Editor, Nature Microbiology.**